Package ‘CAnetwork’

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Title Ecological Networks Analysis with Correspondence Analysis

Version 1.2.0

Description Analyze trait matching and interaction niches in ecological networks using correspondence analyses.

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adegraphics,
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ggplot2,
ggrepel,
grid,
mvtnorm,
patchwork,
rphylopic,
stats,
stringr,
tibble,
tidyr,
tidyselect

Suggests svglite

Contents

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compas2d *Simulate a community matrix*

Description

Simulate abundance values based on species traits. The ecological interpretation can be a sites x species matrix or a resource x consumer species interaction matrix.

Usage

```
compas2d(
  nsp = 40,
  le.grad = 100,
  ratio.grad = 0.8,
  nsite = 100,
  col_abund = NULL,
  row_abund = NULL,
  col_prefix = "s",
  row_prefix = "e",
  rowvar_prefix = "env",
  remove_zeroes = FALSE,
  buffer = 1,
  min.ab = 1,
  max.ab = 100,
  ninter = 100,
  delta = 1,
  mean.tol = 2,
  sd.tol = 10,
  return_intermediate = FALSE
)
```

Arguments

`nsp`	Number of species / consumer
`le.grad`	gradient length for traits/environmental variables. The values are comprised between 0 and `le.grad`.
`ratio.grad`	length of the second gradient as a ratio of the first one.
`nsite`	Number of sites / resource species
`col_abund`	Optional vector of abundances for species/consumers. If missing, abundances are taken from a uniform law between `min.ab` and `max.ab`.
`row_abund`	Optional vector of abundances in sites/ for resource species. If missing, abundances are taken from a uniform law between `min.ab` and `max.ab`.
`col_prefix`	Prefix for the column names of the matrix. Defaults to "s" for "species".
`row_prefix`	Prefix for the row names of the matrix. Defaults to "e" for "environment".

rowvar_prefix
: Prefix for the column names of the row variable. Defaults to "env" for "environment".

remove_zeroes
: If there are unobserved species, should we keep them in the final matrix?

buffer
: buffer to allow species / consumer species optima to fall outside the gradient by 0 - buffer ; le.grad + buffer.

min.ab
: minimal abundance value for the uniform law (if col_abund and/or row_abund are absent)

max.ab
: maximal abundance value for the uniform law (if col_abund and/or row_abund are absent)

ninter
: Total number of observations in the web

delta
: Exponent between 0 and 1 to give more (1) or less (0) weight to trait matching relatively to abundance.

mean.tol
: Mean value for the species/consumers niche tolerance (standard deviation). The species/consumers niche tolerance are drawn from a normal law of mean mean.tol and standard deviation sd.tol for each trait axis.

sd.tol
: Standard deviation for the species/consumers niche tolerance (standard deviation) The species/consumers niche tolerance are drawn from a normal law of mean mean.tol and standard deviation sd.tol for each trait axis.

return_intermediate
: Return intermediate results? If yes, will return the probability of interaction based on matching only, on abundances only and the predicted mix of matching and abundances (before sampling).

Value

An object of class compas, which is a list with 3 components (4 if return_intermediate):

- comm: the community matrix of dimension nsite x nsp. Row names are row_prefix followed by 1..nsite and column names are col_prefix followed by 1..nsp.
- rowvar: the values of the row variables (resource traits/environment gradient values). It is a nsite x 2 matrix. Row names are row_prefix followed by 1..nsite and column names are rowvar_prefix followed by 1 and 2.
- colvar: the values of the column variables (consumer/species traits values). It is a nsp x 2 x 2 array. Row names are row_prefix followed by 1..nsp. Column names in each array dimension are mean and sd (for the mean and standard deviation of the species niche, respectively). Each array 3rd dimension corresponds to a resource trait or environmental gradient value.
- if return_intermediate, a fourth component named intermediate is returned. The first element is named p_matching and contains the matrix of probabilities of interactions based only on matching. The second ab_neutral contains the matrix of count of interactions as predicted by species abundances. The third p_mix contains the matrix of probabilities of interactions taking into account the mix of matching and abundances.

create_code — *Create code*

Description

Create a code for each (unique) value of a vector

Usage

```
create_code(col, code_pattern = NA, name = NA)
```

Arguments

col	column to code
code_pattern	first letter to add before code
name	optional name for output df

Value

dataframe with 2 columns: one with original vector, the other with the corresponding codes

df_to_matrix — *Dataframe to matrix*

Description

Transforms a dataframe into an incidence matrix.

Usage

```
df_to_matrix(df, rows = "plant_species_code", columns = "animal_species_code")
```

Arguments

df	the dataframe to transform. Must have column names indicated in `rows` and `columns` and a third column `frequency` with the interaction frequency/strength index.
rows	Name of the column with the species names that will be the rows of the matrix.
columns	Name of the column with the species names that will be the columns of the matrix.

Value

Returns an incidence matrix with n rows and m columns filled with the values in `frequency`.

`filter_matrix` *Filter matrix*

Description

Filter incidence matrix to keep only species that interact with more partners that a given threshold.

Usage

```
filter_matrix(mat, thr)
```

Arguments

`mat`	The incidence matrix.
`thr`	The minimal number of different interacting partners a species must have (a species must have `>= thr` different interacting partners). It is not weighted (for instance, if `thr` = 2 then a species interacting 100 times with a unique other partner will be removed.)

Value

The interaction matrix but where the rows and/or columns with species that do not interact above the threshold removed.

`format_fit` *Format fits*

Description

Format fits matrices for plotting, i.e. pivot longer each matrix and join them. Optionnally also add bird codes.

Usage

```
format_fit(matx, maty, idcols, bcodes = NULL)
```

Arguments

`matx`	first matrix (values will be in values_x)
`maty`	second matrix (values will be in values_x)
`idcols`	columns to pivor
`bcodes`	optional df of bird codes + names to merge names (codes must be named "bird_species_code")

Value

A long dataframe merged version of the 2 matrices, where each row is an interaction and values of the first and second matrix are respectively in "values_x" and "values_y" multiplot *Plot bi, tri or quadriplots*

Description

Bi, tri or quadriplots from a multivariate analysis

Usage

```
multiplot(
  indiv_row = NULL,
  indiv_col = NULL,
  indiv_row_lab = rownames(indiv_row),
  indiv_col_lab = rownames(indiv_col),
  var_row = NULL,
  var_col = NULL,
  var_row_lab = rownames(var_row),
  var_col_lab = rownames(var_col),
  row_color = "black",
  col_color = "black",
  eig = NULL,
  x = 1,
  y = 2,
  xlim = NULL,
  ylim = NULL,
  grad = 4,
  mult = 1,
  title = NULL,
  max.overlaps = 20,
  force_pull = 1,
  force = 1,
  labsize = 3,
  alphapoints = 0.8
)
```

Arguments

`indiv_row`	Matrix of individuals coordinates for rows (type `dudi$li`)
`indiv_col`	Matrix of individuals coordinates for columns (type `dudi$co`)
`indiv_row_lab`	Labels for row individuals
`indiv_col_lab`	Labels for columns individuals
`var_row`	Matrix of variables coordinates for rows (type `dudi$corR`)
`var_col`	Matrix of variables coordinates for columns (type `dudi$corQ`)
`var_row_lab`	Labels for row variables
`var_col_lab`	Labels for columns variables
`row_color`	Color for the row individuals or variables
`col_color`	Color for the columns individuals or variables
`eig`	Eigenvalues vector

x	Axis to use for the x-axis
y	Axis to use for the y-axis
xlim	x-axis limits. Defaults to data range if not specified
ylim	y-axis limits. Defaults to data range if not specified
grad	graduations for the major.grid
mult	factor to multiply the vectors (arrows on the plot)
title	plot title
max.overlaps	`max.overlaps` argument for `ggrepel::geom_text_repel`
force_pull	`force_pull` argument for `ggrepel::geom_text_repel`
force	`force` argument for `ggrepel::geom_text_repel`
labsize	text size for point labels
alphapoints	Transparency value for points

Value

a ggplot

`patch_plot` *Patch plot*

Description

Arrange plots with patchwork

Usage

```
patch_plot(
  p_data,
  p_resource,
  p_consumer,
  n_consumer,
  n_resource,
  traits_dim = 1
)
```

Arguments

p_data	data matrix plot (generated with `plot_data()`)
p_resource	resource traits plot (generated with `plot_traits()`)
p_consumer	consumer traits plot (generated with `plot_traits()`)
n_consumer	number of resources for plot height
n_resource	number of consumers for plot width
traits_dim	width/height of the traits matrix (use 1 for ordination, 2 for data)

Value

A patchwork ggplot

plotmat *Plot matrices for the model*

Description

Function to plot matrices Y, E (environment/resource traits) and T (species traits/consumer traits)/ This function plots the matrices arranged in a square, with the user-given proportions.

Usage

```
plotmat(
  mar = 2,
  marl = 1.2,
  E = FALSE,
  T_ = FALSE,
  r,
  c,
  l = NULL,
  k = NULL,
  name_mat = TRUE,
  name_dim = TRUE,
  Yname = "italic(Y)",
  Ename = "italic(E)",
  Tname = "italic(T)",
  rname = "italic(r)",
  cname = "italic(c)",
  horizr = TRUE,
  plot.margin.x = 1,
  plot.margin.y = 1
)
```

Arguments

mar	horizontal margin between matrices
marl	vertical margin between matrices
E	Plot matrix E (rows variables)?
T_	Plot matrix T (columns variables)?
r	number of rows of matrix Y
c	number of columns of matrix Y
l	number of columns of matrix E
k	number of columns of matrix T
name_mat	display matrices names?
name_dim	display dimension names?
Yname	name of the matrix Y (center matrix)
Ename	name of the matrix E (row variables matrix)
Tname	name of the matrix T (column variables matrix)
rname	name of the dimension r

cname	name of the dimension c
horizr	display r dimension horizontally? (else, will be vertical)
plot.margin.x	horizontal margin around plot
plot.margin.y	vertical margin around plot

Value

a ggplot of matrix Y (and optionally E and T), with their dimensions written above and aligned in a square where the fourth-corner is missing.

Examples

```
plotmat(r = 5, c = 2)
```

`plot_corcircle` *Plot correlation circle*

Description

Plot the correlation circle

Usage

```
plot_corcircle(
  cor,
  cor2 = NULL,
  xax = 1,
  yax = 2,
  xlab = NULL,
  ylab = NULL,
  label = rownames(cor),
  eig = NULL,
  mar = 0.01,
  labsize = 3.88,
  col_bg = "grey30",
  col = "black",
  lty = "solid",
  col2 = "black",
  lty2 = "solid",
  label2 = rownames(cor2),
  lim = 1,
  clip = "off"
)
```

Arguments

cor	dataframe containing the coordinates of the variables for each axis in columns
cor2	optional second set of variables (for double constrained analyses)
xax	index of the column of the matrix to use for x
yax	index of the column of the matrix to use for y

xlab	Custom x-label
ylab	Custom y-label
label	label to display on the arrows
eig	Eigenvalues vector
mar	margin between arrow tips and label
labsize	Size for arrow labels.
col_bg	color to use for background elements (circle and y = 0 and x = 0)
col	color for arrows (unique)
lty	linetype for the first set of variables
col2	optional color for the second set of variables
lty2	linetype for the second set of variables
label2	optional label to display on the arrows for the second set of variables
lim	limits for x and y: a unique number that gives the distance between the plot limit and zero (so that the circle is centered)
clip	constrain the labels to be inside the plot? If yes, change to "on".

Value

A ggplot with the correlation circle where variables are represented as arrows inside the circle of radius 1.

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plot_eig *Plot eigenvalues*

Description

Barplot of eigenvalues

Usage

```
plot_eig(eigenvalues, showrank = FALSE)
```

Arguments

eigenvalues	The eigenvalues
showrank	Should the x-axis display the rank of the eigenvalues?

Value

A ggplot

plot_table *Plot table*

Description

Plot a matrix

Usage

```
plot_table(
  x,
  y,
  abundance_data,
  abund_table,
  label_consumer = NULL,
  img_consumer = NULL,
  WA = FALSE
)
```

Arguments

`x`	consumers abundance vector
`y`	resources abundance vector
`abundance_data`	abundance data table: expected to have columns
`abund_table`	table corresponding to abundance_data (to wide format = interaction matrix) Consumer_sp, Resource_sp, Abundance and coa_c and coa_r of WA is TRUE.
`label_consumer`	label of consumer species (only useful if WA is TRUE)
`img_consumer`	image of consumer species (only useful if WA is TRUE)
`WA`	display the weighted averaging for one species?

Value

a ggplot

plot_traits *Plot traits*

Description

Plot trait matrices

Usage

```
plot_traits(traits, order, label, img, fill = "grey", type = "consumer")
```

Arguments

`traits`	traits dataframe (expected to have a `size` column)
`order`	order to use for the species
`label`	species labels
`img`	species image
`fill`	fill color (use a unique color to ignore the size)
`type`	either "resource" (row) or "consumer" (column)

Value

a ggplot

`plot_traits_data` *Plot traits data*

Description

Plot trait data matrices

Usage

```
plot_traits_data(traits, label, img, fill, type = "consumer", reorder = FALSE)
```

Arguments

`traits`	traits dataframe (expected to have a `size` column)
`label`	species labels
`img`	species image
`fill`	fill color (use a unique color to ignore the size)
`type`	either "resource" (row) or "consumer" (column)
`reorder`	should species be reordered by label?

Value

a ggplot

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